Learning Energy-Efficient Uplink Access with Asynchronous Packet Deadlines

Anonymous Authors
Complete first draft — author information to be supplied

***Abstract*—Periodic uplink devices can save transmit energy by using lower power over several slots, but overlapping transmissions can erase this benefit when devices independently select their resources. We study distributed subchannel, power, and silence decisions for heterogeneous periodic packets with asynchronous arrival phases and per-device deadline-violation requirements. Each device uses its own packet state, delayed public received-power measurements, and private mutual-information feedback; it observes neither other devices' actions nor their queues. A shared recurrent policy is trained with multi-agent proximal policy optimization and evaluated against calibrated local rules and centralized zero-loss energy references. On 200 held-out topology–phase instances per network size, with three training seeds and two executions per instance, the fixed-penalty policy reduces mean normalized transmit energy by 36.6%, 56.7%, and 70.5% relative to the strongest evaluated feasible local rule for 6, 8, and 12 devices sharing four subchannels. Its feasible-episode fraction is at least 0.984 at these loads. With four devices, its energy is within 0.7% of the exact zero-loss centralized reference. Validation diagnostics associate the savings with reduced co-channel overlap and interference loss. Local dual adaptation gives no statistically resolved energy advantage over the fixed penalty, while substantial training remains necessary. These results quantify the benefits and remaining limits of learned access under the specified local information model.**

***Index Terms*—distributed access, periodic traffic, deadline constraints, power control, multi-agent reinforcement learning**

I. Introduction

Periodic sensing and control traffic poses an energy–reliability tradeoff in shared uplinks. A device can lower its transmit power and accumulate service across several slots before its deadline. However, extending transmission also increases the opportunities to interfere with other devices. With heterogeneous periods and unaligned arrival phases, independently choosing a low power does not ensure low network energy. Devices must also learn when to transmit and which subchannel to use, including when to remain silent despite a pending packet.

The energy benefit of stretching a transmission under deadline constraints is established by classical packet-scheduling results [1]. In a shared uplink, the practical question is how much of this benefit a distributed controller can retain using delayed local feedback. High packet delivery alone does not answer this question: frequent maximum-power transmissions may be reliable but expensive. Conversely, low energy can result from leaving difficult devices unserved. A useful evaluation must therefore combine per-device reliability checks, an energy-aware local baseline, and an explicit centralized reference.

Learning-based distributed spectrum access has been studied using recurrent policies with local acknowledgments [2]. Joint subband and power decisions for delivering a payload within a time budget appear in vehicular multi-agent reinforcement learning (MARL) [3]. For periodic mission-critical traffic, reinforcement-learning power control has also been developed for reliability-constrained single-link transmission [4]. More recently, recurrent MARL has addressed spectrum and power allocation with stochastic traffic in multicell networks [5]. These works establish that neither joint resource selection nor recurrent MARL is by itself a new contribution.

The particularly close XR-SBPC study [6] combines periodic traffic with offsets, accumulated Shannon service, and discrete subband–power actions, including silence. It addresses reliability in a network of downlink subnetworks and uses observations that include neighboring actions and current link gains. Our study instead examines transmit-energy minimization under per-device delivery requirements in a common-receiver uplink. Execution uses only the device's own history and AP feedback, without access to other devices' actions, queues, or phases. We evaluate this information model against centralized references computed for the same packets and discrete power menu.

We instantiate a shared, subchannel-equivariant recurrent policy within multi-agent proximal policy optimization (MAPPO) [7], and compare a fixed violation penalty with local dual adaptation inspired by state-augmented constrained reinforcement learning [8]. The contribution is an evaluated access-control formulation and benchmark, with three concrete components:

1) We specify asynchronous periodic delivery with accumulated service, joint subchannel–power–silence actions, and an explicit causal information boundary for decentralized execution.
2) We construct discrete-power zero-loss energy references: an exact interference-free solution when devices do not outnumber subchannels, and a separable lower reference plus a feasible cyclic orthogonal schedule at higher loads.
3) We quantify held-out energy and reliability with multiple training seeds, calibrated local rules, and paired uncertainty estimates. Additional diagnostics identify reduced transmission overlap as an observed correlate of the

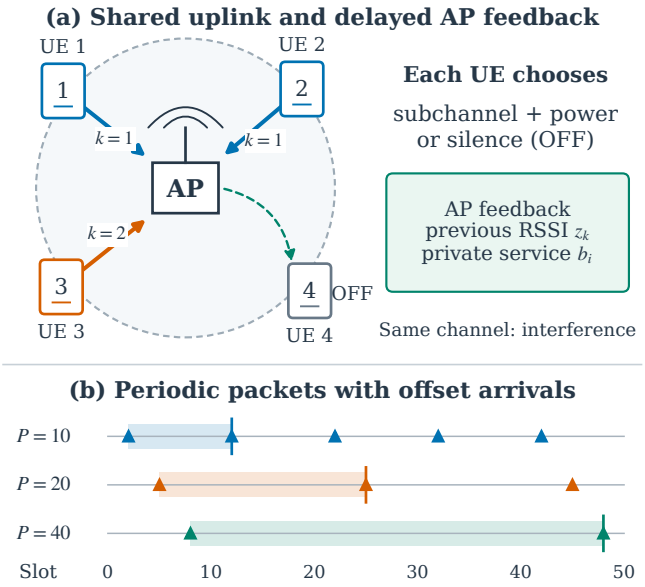


Fig. 1. Periodic uplink access. Solid arrows illustrate transmissions; UEs 1 and 2 reuse a subchannel and interfere at the AP. Dashed feedback is received even during silence. Shading marks the first packet window in each illustrative traffic class; triangles denote arrivals and vertical bars its deadline. Physical slots are synchronized, while arrival phases differ.

gains and expose the cost of insufficient training and residual per-device failures.

Our conclusions concern static, frequency-flat channels and transmit energy. They establish neither a general constraint guarantee for MAPPO nor superiority over all available learning algorithms.

II. System Model and Control Objective

A. Periodic uplink and accumulated service

Figure 1 illustrates an access point (AP) receiving uplink packets from N devices over K orthogonal subchannels. Slots have duration τ, and each subchannel has bandwidth B. Device i generates a packet of L_i bits at $a_{i,m} = \varphi_i + mP_i$, with deadline $d_{i,m} = a_{i,m} + P_i$. The phase φ_i is drawn independently and uniformly from $\{0, \dots, P_i - 1\}$. Physical slots are synchronized; asynchronous traffic refers to the packet-arrival phases. At most one packet is pending per device.

In slot t, device i selects $a_i(t) \in \{\mathsf{OFF}\} \cup (\{1, \dots, K\} \times \mathcal{P})$, where $\mathcal{P}$ is the positive discrete-power menu. Let $k_i(t)$ and $p_i(t)$ denote its chosen subchannel and power, with $p_i(t) = 0$ for OFF. The AP treats interference as noise, without successive interference cancellation. For an active device,

$$\gamma_i(t) = \frac{p_i(t) g_i}{\sigma^2 + \sum_{j \neq i : k_j(t) = k_i(t)} p_j(t) g_j}, \tag{1}$$

$$b_i(t) = \tau B \log_2(1 + \gamma_i(t)), \qquad e_i(t) = \tau p_i(t). \tag{2}$$

The gain g_i is static and identical across subchannels within an instance; thus subchannel choice controls interference rather than frequency diversity. For silence, $b_i(t) = e_i(t) = 0$.

We use an ideal accumulated-mutual-information model, motivated by incremental redundancy hybrid ARQ [9]. A packet completes once the sum of its slot services reaches L_i. This is an information-theoretic abstraction, with no finite-blocklength penalty or coding gap. Even if its final transmission provides excess service, a device occupies and pays for the whole slot. At each slot boundary, the environment first applies service, then settles expiring packets, and finally releases new packets. Unfinished bits are discarded at expiry. A device without a pending packet must select OFF but continues receiving observations. Only radiated transmit energy is counted; circuitry, feedback, and processing energy are excluded.

B. Causal execution information

At a slot's start, device i knows its remaining bits $s_i(t)$, remaining time $D_i(t)$, its packet size and period, and its preceding action and outcome. The AP supplies the previous slot's per-subchannel received power $z_k(t-1)$, including noise, and the device's private service feedback $b_i(t-1)$. It supplies no scheduling decision. Devices do not exchange messages, and the policy receives no device identifier, global slot index, topology, or other devices' queues, phases, gains, and actions.

These observations contain useful channel information. After transmitting at power p on subchannel k, a device can infer

$$\widehat{\gamma} = 2^{b/(\tau B)} - 1, \qquad \widehat{I} = \frac{z_k}{1 + \widehat{\gamma}}, \qquad \widehat{g}_i = \frac{z_k - \widehat{I}}{p}. \tag{3}$$

The gain estimate is retained across packets. Exact feedback and a static channel make this estimate exact after a transmission in the ideal model; the remaining uncertainty includes other devices' future actions. All distributed baselines use the same permitted observation stream.

C. Energy and per-device reliability

Let $c_{i,m} = 1$ if packet m of device i expires unfinished. For a fixed instance, the intended policy objective is

$$\min_{\pi} \quad J(\pi) = \limsup_{T \to \infty} \frac{1}{NT} \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \sum_i e_i(t) \right], \tag{4}$$

$$\text{s.t.} \quad \limsup_{M \to \infty} \frac{1}{M} \mathbb{E}_\pi \left[\sum_{m=0}^{M-1} c_{i,m} \right] \leq \epsilon, \quad \forall i.$$

The policy is trained across random instances and must act from each device's local history. The per-device constraints prevent good aggregate delivery from hiding a persistently underserved device. We use a penalized learning surrogate and assess reliability empirically on finite episodes; our evaluation criterion in Section V is not a proof that (4) holds for every instance.

III. Centralized Energy References

We compute reference energies with full instance information and the same power menu, slot duration, and service model. These references require *every packet* to meet its deadline. The subscript 0 below denotes this zero-loss requirement, distinct from the $\epsilon = 0.05$ allowance for learned policies.

A. Interference-free per-packet solution

For device i, define the isolated service and energy of level ℓ as $b_{i\ell} = \tau B \log_2(1 + p_\ell g_i/\sigma^2)$ and $e_\ell = \tau p_\ell$. Its minimum isolated packet energy is the small integer problem

$$q_i = \min_{\boldsymbol{n} \in \mathbb{Z}_+^{|\mathcal{P}|}} \left\{ \sum_\ell n_\ell e_\ell : \sum_\ell n_\ell b_{i\ell} \geq L_i, \quad \sum_\ell n_\ell \leq P_i \right\}. \tag{5}$$

Static channels make transmission slots interchangeable for this isolated problem. We enumerate the counts of all but the lowest positive power and fill the lowest level minimally. With five levels and periods no larger than 40, this yields an exact discrete solution. For a fixed evaluated packet set with M_i packets from device i, let $L_0 = \sum_i M_i q_i$.

Proposition 1. *For the zero-loss centralized scheduling problem, L_0 is a lower bound on energy. If $N \leq K$, assigning one subchannel to each device attains L_0.*

Proof. For fixed power, service in (1) cannot exceed isolated service. Every successfully delivered packet therefore costs at least q_i; summing over packets gives the lower bound. If $N \leq K$, distinct subchannels eliminate interference. Repeating each device's solution of (5) within its nonoverlapping packet windows attains the sum, independently of arrival phases. □

B. Cyclic orthogonal witness at higher loads

Let $C = \mathrm{lcm}(P_1, \ldots, P_N)$, which is 40 slots in our setup. For every cyclic packet window w of device $i(w)$, choose integer power-level counts $n_{w\ell}$. Let $y_{it} \in [0, 1]$ indicate slot assignment over one cycle. An orthogonal schedule is obtained from

$$\begin{aligned} \min_{n,y} \quad & \sum_{w,\ell} e_\ell n_{w\ell} \\ \text{s.t.} \quad & \sum_{t \in w} y_{i(w)t} = \sum_\ell n_{w\ell}, \quad \forall w, \\ & \sum_i y_{it} \leq K, \quad \forall t, \\ & \sum_\ell b_{i(w)\ell} n_{w\ell} \geq L_{i(w)}, \quad \forall w, \\ & n_{w\ell} \in \mathbb{Z}_+, \quad 0 \leq y_{it} \leq 1. \end{aligned} \tag{6}$$

For fixed integer counts, the assignment constraints form a capacitated bipartite flow problem with integral capacities. An integral assignment can therefore be recovered by maximum flow. Each slot's active devices are assigned distinct subchannels, and the selected powers are placed within their packet windows. Repeating this schedule gives a feasible zero-loss witness, whose energy U_0 is measured by replay in the communication environment. We verify packet completion and energy accounting for every reference instance.

Consequently $L_0 \leq E_0^\star \leq U_0$ for the same zero-loss packet set, with equality when $N \leq K$. Orthogonality is only a restriction for $N > K$; we make no optimality claim for it there. We report learned packet energy normalized by both references. Since learned policies may drop packets, E/L_0 and E/U_0 characterize their energy relative to zero-loss references, not a certified approximation factor for the constrained objective (4). In particular, proximity to the $N = 4$ reference does not establish optimality when a 5% violation allowance can be used deliberately.

IV. Learning Distributed Access

A. Shared recurrent actor

Figure 2 shows the actor and centralized training pathway. All devices share one actor but maintain private recurrent states. Its subchannel tokens contain delayed RSSI, inferred interference, indicators of the device's preceding transmission and power level, and estimated achievable service at each power divided by required service s_i/D_i. The context contains normalized remaining bits and time, the period, previous outcome and service, and the remembered gain with a validity indicator. Logarithmic scaling and clipping are applied to the rate ratios and physical quantities.

A shared encoder and gated recurrent unit (GRU) process each subchannel token $x_{ik}(t)$ together with device context $u_i(t)$:

$$h_{ik}(t) = \mathrm{GRU}_\theta(f_\theta(x_{ik}(t), u_i(t)), h_{ik}(t-1)), \tag{7}$$

$$v_i(t) = g_\theta\left(K^{-1} \sum_k \phi_\theta(h_{ik}(t)), u_i(t) \right). \tag{8}$$

A shared head maps (h_{ik}, v_i, u_i) to the power logits for subchannel k; a separate head maps (v_i, u_i) to the silence logit. A softmax over the $1+K|\mathcal{P}|$ outputs defines π_θ. This set aggregation follows the permutation-invariant construction of Deep Sets [10]; permuting the subchannel histories permutes their action logits. The GRU state is updated during idle slots and retained across packet boundaries, then reset when a new training or evaluation episode begins. Empty devices are masked to OFF.

B. Training surrogate and constraint variants

A centralized state-value critic observes all devices' features and true gains during training. Let $c_i(t)$ be the deadline-violation indicator settled at the end of slot t. The common training reward is

$$r_t = -\frac{1}{N}\left(\sum_i \frac{e_i(t)}{E_{\mathrm{ref}}} + \sum_i \lambda_i(t) c_i(t) \right), \quad E_{\mathrm{ref}} = \tau p_{\max}. \tag{9}$$

Our primary method, *MAPPO-F*, sets $\lambda_i = w = 10$ for all devices. This fixed penalty is selected on validation data; it is a surrogate for the per-device constrained problem, not an exact constrained solver.

The comparison method, *MAPPO-D*, maintains a multiplier locally at each device. At each of that device's deadlines,

$$\lambda_i \leftarrow \mathrm{clip}\big(\lambda_i + \eta(c_i - \epsilon), 0, \lambda_{\max}\big). \tag{10}$$

The multiplier is also an actor input. We use $\eta = 2$ and $\lambda_{\max} = 50$, independent uniform initial multipliers on $[0, 50]$

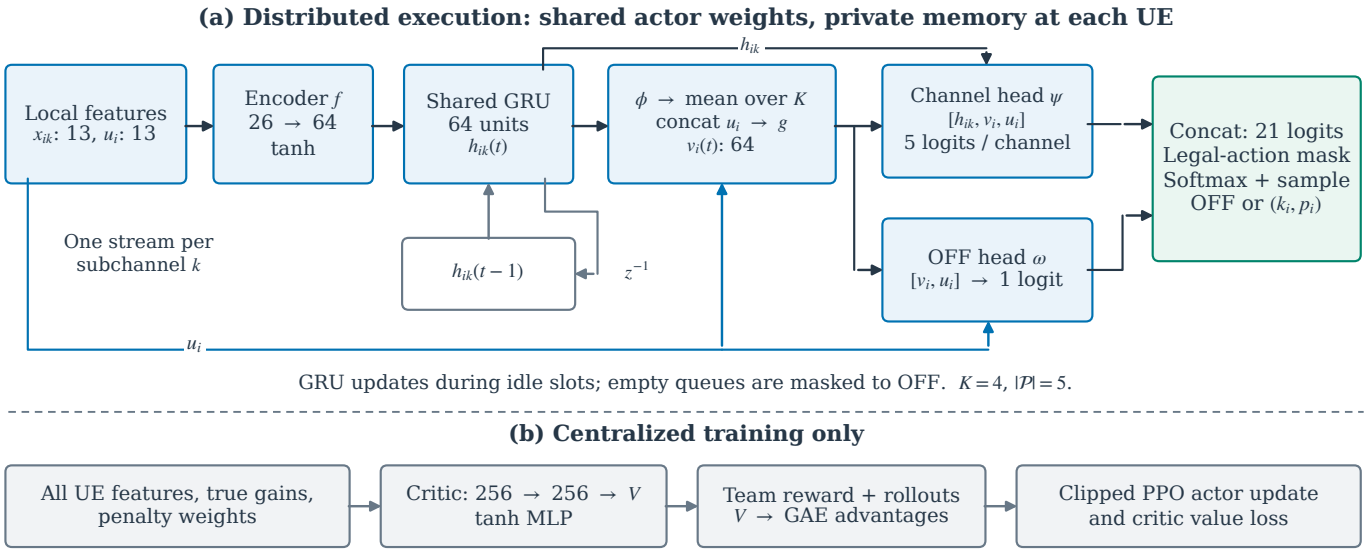


Fig. 2. Recurrent MAPPO architecture. Each UE runs the same actor with private recurrent states, one per subchannel; the encoder, GRU, and channel head share weights over k. Blue routing supplies the local context u_i; both action heads also receive it. The delay z^{-1} retains hidden state across slots. The critic and global training inputs are absent at execution. MAPPO-D adds the local multiplier to the context; MAPPO-F keeps that entry zero.

during training, and a common initial value selected from $\{0, 5, 25\}$ during validation. Execution updates depend only on local outcomes. This is an application of state augmentation with dual dynamics [8], with no claim of a new dual algorithm or inherited convergence guarantee under function approximation.

Both variants use recurrent MAPPO with a common generalized advantage estimate from the centralized critic. For ratio $\rho_{it} = \pi_\theta(a_i(t) \mid h_i(t))/\pi_{\theta_{\text{old}}}(a_i(t) \mid h_i(t))$, the actor maximizes

$$\mathbb{E}\Big[\min\{\rho_{it}\widehat{A}_t, \text{clip}(\rho_{it}, 1-\delta, 1+\delta)\widehat{A}_t\} + \beta\mathcal{H}(\pi_\theta)\Big]. \quad (11)$$

Only devices with pending packets contribute to the policy loss and entropy; the critic learns from all slots. Ordered trajectory replay preserves recurrent state. Technical episode truncations bootstrap the critic rather than marking the continuing communication task terminal.

TABLE I
SCENARIO AND LEARNING PARAMETERS.

Parameter	Value
Devices / subchannels	$N \in \{4, 6, 8, 12\}$ / $K = 4$
Slot duration / bandwidth	1 ms / 1 MHz per subchannel
Cell radius / path-loss exponent	100 m / 4
Noise density / noise figure	-174 dBm/Hz / 7 dB
Positive transmit powers	5, 10, 15, 19, 23 dBm
Periods P_i (slots)	10, 20, 40
Packet sizes L_i (bits)	6144, 12 288, 24 576
Per-device violation allowance	$\epsilon = 0.05$
Evaluation horizon / warm-up	2000 / 200 slots
Training episode / rollout	600 / 128 slots
Parallel environments / epochs	16 / 4 PPO epochs
Actor GRU / critic hidden widths	64 / 256, 256
Actor / critic learning rates	3×10^{-4} / 10^{-3}
Discount / GAE parameter	0.99 / 0.95
PPO clip / gradient norm limit	0.2 / 0.5
Training budget per seed	20×10^{6} environment slots

V. Experimental Evaluation

A. Scenario, baselines, and protocol

Table I gives the common scenario and training parameters. Device positions are sampled uniformly by area in a disk; gains are $g_i = 10^{-4}\max(d_i/1\,\text{m}, 1)^{-4}$. Positions and phases remain fixed within an episode. The three traffic classes have equal offered load per device, so increasing N raises load without changing its per-device average. The class counts for $N = 4, 6, 8, 12$ are respectively $(2, 1, 1)$, $(2, 2, 2)$, $(3, 3, 2)$, and $(4, 4, 4)$.

We train a separate shared policy for each N, using seeds 101, 202, and 303. Each episode draws a topology–phase instance uniformly from a pool of one million training instances. Training, development, validation, and test pools are disjoint. Actor and critic parameters are not shared across network sizes; the study tests generalization to new instances at a trained N, rather than zero-shot transfer to a new N. The entropy coefficient decreases linearly from 0.01 to zero over the first ten million slots. Total training slots are summed over the 16 environments. Minibatches contain four environment trajectories. The penalty and dual step size were selected in an initial two-million-slot study at $N = 8$ and transferred to all long runs.

Four local rules provide non-learning comparisons. *Random-max* selects a random subchannel and maximum power whenever a packet is pending. *Sticky-max* retains its preceding subchannel if its previous service meets the current required pace, and otherwise reselects randomly, always at maximum power. *Lazy-pacing* selects power from its latest link estimate to meet s_i/D_i, changing subchannel when

TABLE II
ALL DISTRIBUTED METHODS ON THE HELD-OUT TEST SET. EACH CELL IS F / MEAN E/U_0. † DENOTES FAILURE OF THE $F \geq 0.95$ CRITERION; ITS ENERGY IS DESCRIPTIVE ONLY.

Method	$N = 4$	$N = 6$	$N = 8$	$N = 12$
MAPPO-F	0.993/1.01	0.986/1.09	0.994/1.21	0.984/1.62
MAPPO-D	0.992/1.01	0.993/1.11	0.979/1.46	0.972/1.52
Paced-sticky	1.000/1.00	0.990/1.73	0.960/2.80	0.980/5.51
Random-max	1.000/9.87	1.000/9.32	1.000/9.34	0.995/8.93
Sticky-max	1.000/10.36	1.000/10.05	1.000/10.40	0.983/10.21
Lazy-pacing	0.797/5.34†	0.657/4.98†	0.287/5.76†	0.000/8.02†

necessary. The strongest evaluated feasible rule, *paced-sticky*, remembers its gain across packets, retains a subchannel within 3 dB of the least-interfered one, and otherwise switches with probability 0.5 to a subchannel within 3 dB of the minimum. It chooses the lowest power predicted to serve $(1 + m)s_i/D_i$ bits, probing at maximum power before its gain is known. Its margin m is calibrated over $\{0, 0.1, 0.25, 0.5, 1, 2, 4, 8, 16, 32\}$ separately for each N, selecting $0, 0.25, 0.5, 8$, respectively. Thus it is an energy-aware, load-calibrated comparison rather than only a maximum-power random-access rule.

Validation uses 64 instances and one stochastic execution per instance, approximately every million training slots. An episode is feasible if every device has empirical violation rate at most 0.05. With M_i evaluated packets and V_i violations, this is exactly $V_i \leq \lfloor 0.05 M_i \rfloor$ for all i. We denote the fraction of feasible episodes by F. Among checkpoints with validation $F \geq 0.95$, we select the smallest mean normalized energy; if none qualifies, we select the highest F, breaking ties by energy. MAPPO-D jointly selects its initial multiplier. The selected and final checkpoints are both evaluated.

Testing uses 200 held-out instances per N, two executions per instance, and all three training seeds. Random execution uses independent private device streams with common seed assignments across methods. Evaluated packets have deadlines in $(200, 2000]$; their complete transmit energy, including energy spent on failed packets, defines E. This packet-set accounting matches the centralized denominators. Separately, reported mean power counts all transmissions in slots $[200, 2000)$. Methods satisfy the empirical protocol criterion when their aggregate $F \geq 0.95$. This threshold uses the point estimate, not the lower confidence limit. Energy superiority is judged only between methods satisfying it. We use 10 000 two-level bootstrap resamples over training seeds and instances, averaging repetitions within an instance. Differences resample instances jointly and training seeds independently between methods. All intervals are 95% percentile intervals.

B. Held-out energy and reliability

Figure 3 and Table II show that MAPPO-F satisfies the empirical feasibility criterion at all four loads. For $N = 6, 8, 12$, its mean E/U_0 is 1.094, 1.213, and 1.623, compared with 1.725, 2.798, and 5.508 for paced-sticky. The reductions are 36.6%, 56.7%, and 70.5%, computed as one minus the ratio of the two methods' mean normalized energies. They are not means of per-instance percentage reductions. The paired differences (MAPPO-F minus paced-sticky) have intervals $[-0.659, -0.603]$, $[-1.632, -1.541]$, and $[-4.179, -3.535]$, respectively; all exclude zero. Corresponding mean transmit powers are 4.36, 5.30, and 8.87 mW/device, versus 6.83, 12.07, and 29.24 mW/device for paced-sticky.

At $N = 4$, paced-sticky attains the exact zero-loss reference. MAPPO-F has $E/U_0 = 1.007$ with interval $[1.004, 1.010]$, and $F = 0.993$. This is a useful reference-recovery check, with no improvement over the local rule at this load. At higher loads the learned policy's advantage over the local rules grows, yet its distance from centralized scheduling also grows. MAPPO-F's mean E/L_0 is 1.119, 1.349, and 2.289 for $N = 6, 8, 12$. At $N = 12$, energy remains 62.3% above the feasible orthogonal reference. These ratios show substantial remaining room for improvement, with the zero-loss qualification of Section III.

Maximum-power rules retain high feasibility but consume approximately 9–10 times the centralized reference energy. Lazy-pacing fails the feasibility criterion at every load, reaching $F = 0$ at $N = 12$. Thus lowering power using only a recent pacing estimate does not by itself resolve multiuser interference. The calibrated paced-sticky rule is substantially stronger and remains the main energy comparison.

C. Does local dual adaptation improve performance?

Both selected MAPPO variants satisfy the empirical criterion at every N. However, the paired energy difference between MAPPO-F and MAPPO-D has a 95% interval containing zero at every load. At $N = 8$, their means are 1.213 and 1.463, but the difference interval is $[-0.737, 0.005]$; at $N = 12$, the means are 1.623 and 1.522, with interval $[-0.119, 0.452]$. One less effective training seed strongly affects each comparison. These results give no statistically resolved energy advantage to either constraint treatment; they do not establish equivalence.

Checkpoint dependence provides a practical distinction. MAPPO-F's final checkpoints remain feasible at all four loads, with F between 0.987 and 0.992. MAPPO-D's final F falls to 0.915 and 0.938 at $N = 8$ and 12. The fixed-penalty variant therefore provides the simpler empirical reference for this study. A separate short-budget global-multiplier variant selected its untrained initial checkpoint and achieved test $F = 0.592$ at $N = 8$. That result concerns this training configuration and selection outcome; it does not show that global primal–dual methods generally fail.

D. Training and observed coordination

The gain requires substantial training. Figure 4 shows a large energy decrease between roughly six and twelve million slots. The separately trained two-million-slot MAPPO-F configuration has test $E/U_0 = 2.321$ and $F = 0.989$ at $N = 8$; the long configuration reaches 1.213 and 0.994. Budget and entropy schedule both change between these configurations, so the difference cannot be attributed to either change alone.

To examine behavior, we replay selected policies on the validation set. Co-channel overlap is the fraction of transmissions sharing their subchannel with another active device;

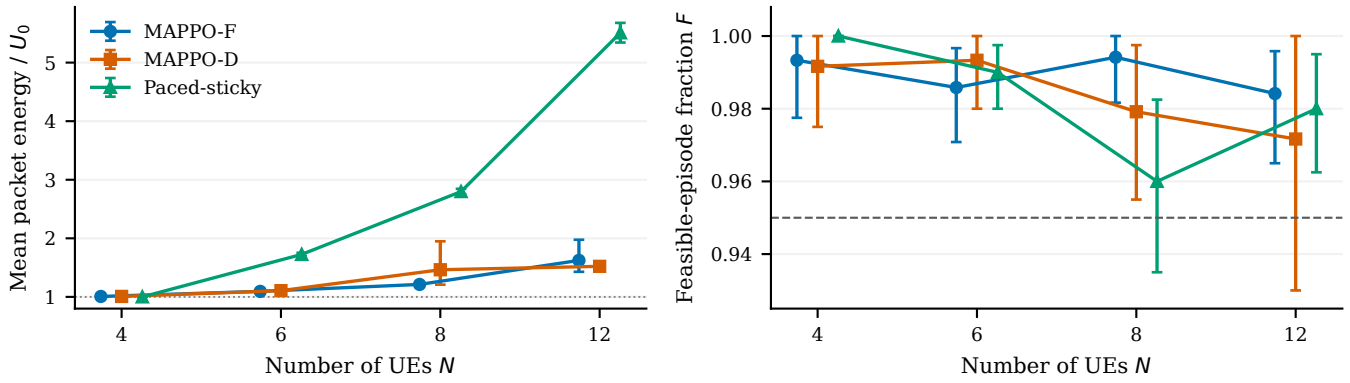


Fig. 3. Held-out test performance of selected checkpoints. The energy denominator U_0 is a feasible centralized zero-loss schedule; it equals the zero-loss optimum for $N = 4$. Bars show two-level 95% bootstrap intervals. The dashed line at $F = 0.95$ is the empirical protocol threshold.

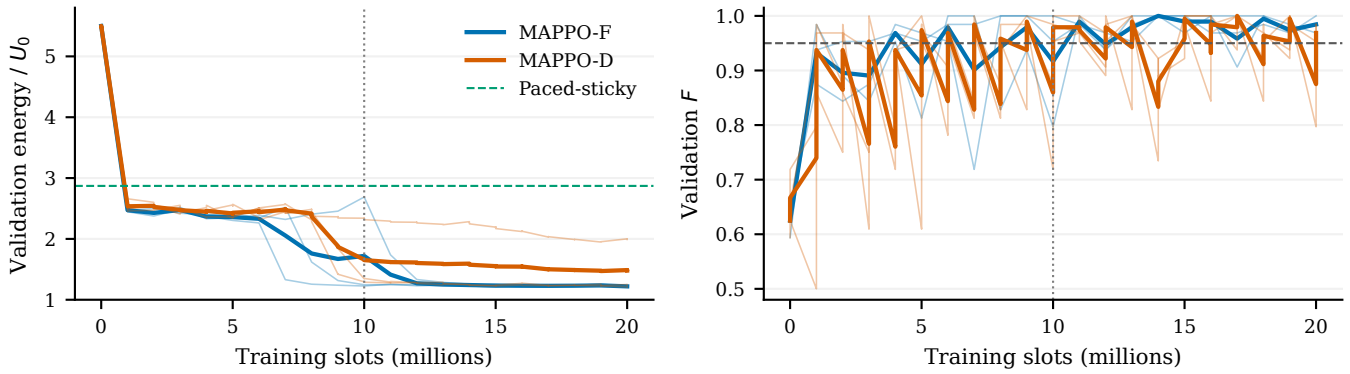


Fig. 4. Validation trajectories for $N = 8$. Thin lines show the three training seeds; thick lines show their means, without smoothing. The vertical dotted line marks the end of entropy annealing at ten million slots. For MAPPO-D, curves use the validation-selected initialization at each checkpoint. These are training diagnostics, not additional test-set evaluations.

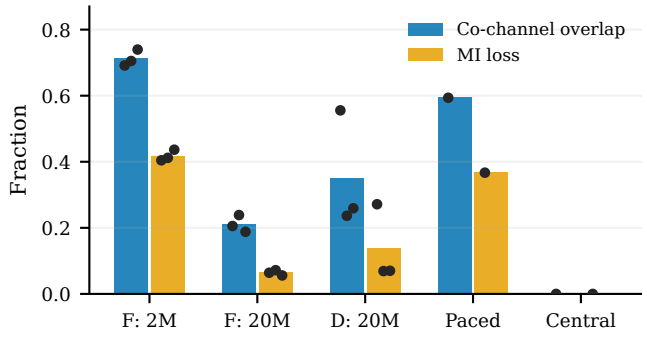


Fig. 5. Descriptive validation diagnostics at $N = 8$: short (2M) and long (20M) training configurations. F and D denote MAPPO-F and MAPPO-D. Bars are means across seeds; dots are individual seed summaries. The central reference and paced-sticky have one summary each.

it does not mean automatic packet loss in the SINR model. Interference loss is one minus the ratio of total actual mutual information to the interference-free mutual information of those same transmissions at the same powers. For MAPPO-F, the short-budget configuration has mean overlap 0.712 and loss 0.418; the long configuration reduces these to approximately 0.211 and 0.064 (Fig. 5). Paced-sticky has overlap 0.594 and loss 0.367. Long-trained MAPPO-F also uses the 5 and 10 dBm levels for about 83% of transmissions. Its probability of transmitting conditional on having a pending packet is 0.61–0.76 across seeds, versus 1 for paced-sticky, consistent with using silence while work is pending. This conditional fraction is not the fraction of all device–slots that are occupied.

Together these observations support an association between energy savings, lower-power transmission, and less harmful overlap. They do not isolate the causal contributions of recurrence, power adaptation, or silence. A separate validation-only short-budget experiment appended interference and own-action features at lags of 10, 20, and 40 slots. It changed MAPPO-F's mean E/U_0 from 2.337 to 2.432, and MAPPO-D's from 2.404 to 2.385; neither met the predeclared 5% improvement threshold. This negative result is limited to the tested features and budget, not all ways of exploiting periodicity.

E. Reliability tails and scope

The aggregate feasibility criterion allows a small set of bad episodes. In a post hoc analysis pooling both learned variants over all four loads, 128 of 9600 executions are infeasible. In 108 of these, the worst device is 75–100 m from the AP, and in 83 its period is 40 slots. Such a device has 45 evaluated packets and can miss only two before the episode becomes infeasible. Some failures are severe: an $N = 4$ MAPPO-F execution contains a device violation rate of 0.733. Average feasibility must therefore not be read as a uniform reliability guarantee or a URLLC claim.

Three training seeds limit precision about optimization variability, and the tested channel and feedback are idealized. Accounting for feedback, circuit, and processing energy could change the benefit of spreading transmissions. Further, this completed evaluation contains no independently implemented literature-based learning comparator; it establishes an advantage over the evaluated local rules. An information-matched XR-SBPC-style SAC comparison is outside the completed evidence reported here.

VI. Conclusion

For the studied periodic uplink, a shared recurrent MAPPO policy learned useful distributed access behavior from delayed AP feedback and private packet state. At loads exceeding the number of subchannels, the fixed-penalty variant substantially reduced transmit energy relative to calibrated local rules while meeting the empirical per-device feasibility criterion. Centralized zero-loss references expose both recovery of the low-load reference and a growing energy gap at higher load. Local dual adaptation did not yield a statistically resolved energy advantage, and observed coordination required long training. Literature-based learning comparisons, controlled mechanism ablations, and robustness to imperfect feedback and time-varying channels are the next steps needed to extend these conclusions.

References

[1] E. Uysal-Biyikoglu, B. Prabhakar, and A. E. Gamal, "Energy-efficient packet transmission over a wireless link," *IEEE/ACM Transactions on Networking*, vol. 10, no. 4, pp. 487–499, 2002.

[2] O. Naparstek and K. Cohen, "Deep multi-user reinforcement learning for distributed dynamic spectrum access," *IEEE Transactions on Wireless Communications*, vol. 18, no. 1, pp. 310–323, 2019. [Online]. Available: https://arxiv.org/abs/1704.02613

[3] L. Liang, H. Ye, and G. Y. Li, "Spectrum sharing in vehicular networks based on multi-agent reinforcement learning," *IEEE Journal on Selected Areas in Communications*, vol. 37, no. 10, pp. 2282–2292, 2019. [Online]. Available: https://arxiv.org/abs/1905.02910

[4] C. Guo, Z. Li, L. Liang, and G. Y. Li, "Reinforcement learning-based power control for reliable mission-critical wireless transmission," *IEEE Internet of Things Journal*, vol. 10, no. 23, pp. 20 868–20 883, 2023. [Online]. Available: https://arxiv.org/abs/2202.06345

[5] Y. Zhang and D. Guo, "Multi-agent reinforcement learning for multi-cell spectrum and power allocation," *IEEE Transactions on Communications*, 2025. [Online]. Available: https://arxiv.org/abs/2312.05746

[6] S. Bagherinejad, T. Jacobsen, N. K. Pratas, and R. O. Adeogun, "DRL-based distributed joint sub-band allocation and power control for eXtended Reality over in-body subnetworks," in *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, 2025. [Online]. Available: https://doi.org/10.1109/WCNC61545.2025.10978802

[7] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," in *Advances in Neural Information Processing Systems*, vol. 35, 2022. [Online]. Available: https://arxiv.org/abs/2103.01955

[8] M. Calvo-Fullana, S. Paternain, L. F. O. Chamon, and A. Ribeiro, "State augmented constrained reinforcement learning: Overcoming the limitations of learning with rewards," 2021, arXiv:2102.11941. [Online]. Available: https://arxiv.org/abs/2102.11941

[9] G. Caire and D. Tuninetti, "The throughput of hybrid-ARQ protocols for the Gaussian collision channel," *IEEE Transactions on Information Theory*, vol. 47, no. 5, 2001. [Online]. Available: https://www.eurecom.fr/publication/394

[10] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. J. Smola, "Deep sets," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://arxiv.org/abs/1703.06114